Distributed Data Processing Environments

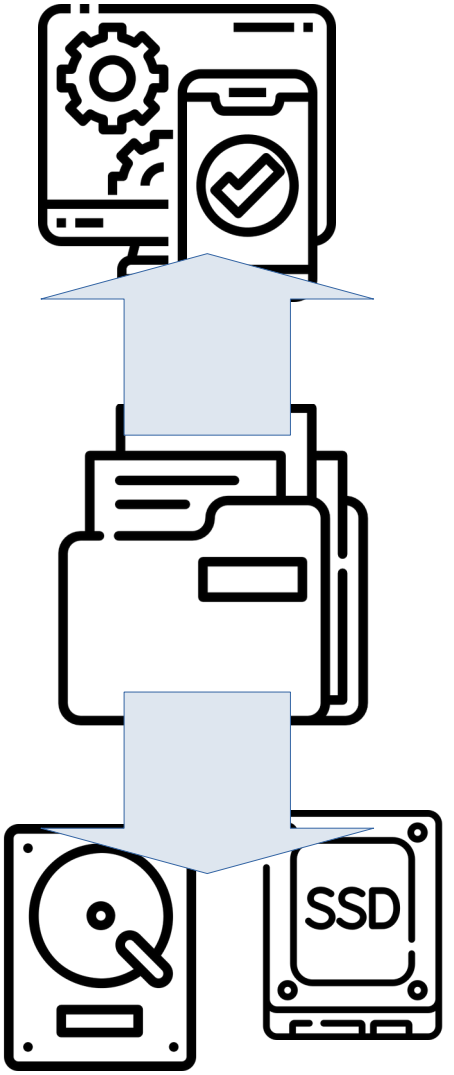
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Storage stack

- What is stored in files?
- How are files stored?



Icons by Flaticon.com.

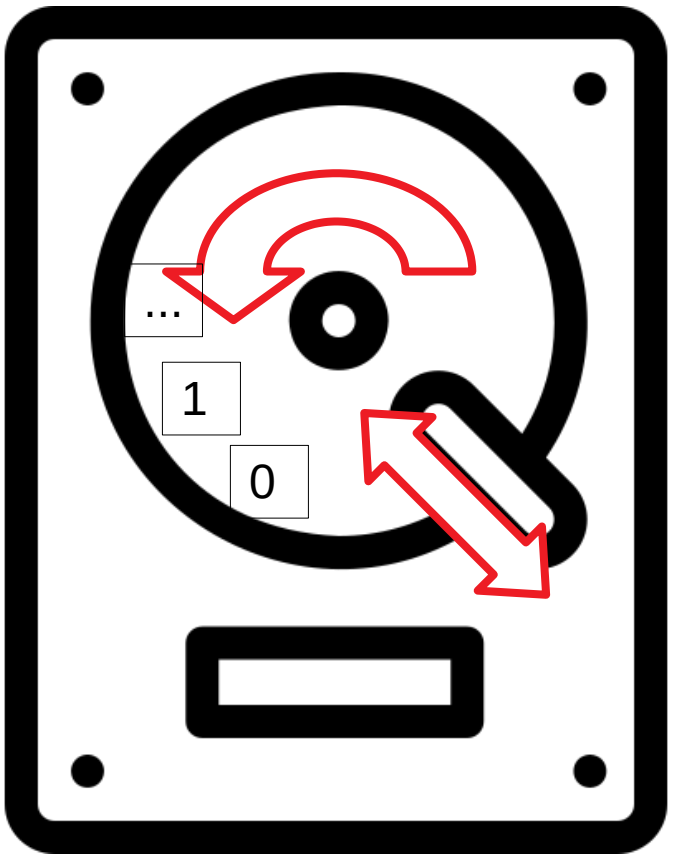
Logical disk

- Sequence of fixed size sectors
  - e.g. 512 bytes
- Available operations:
  - Read sectors(s)
  - Write sectors(s)



Hard disks (HD)

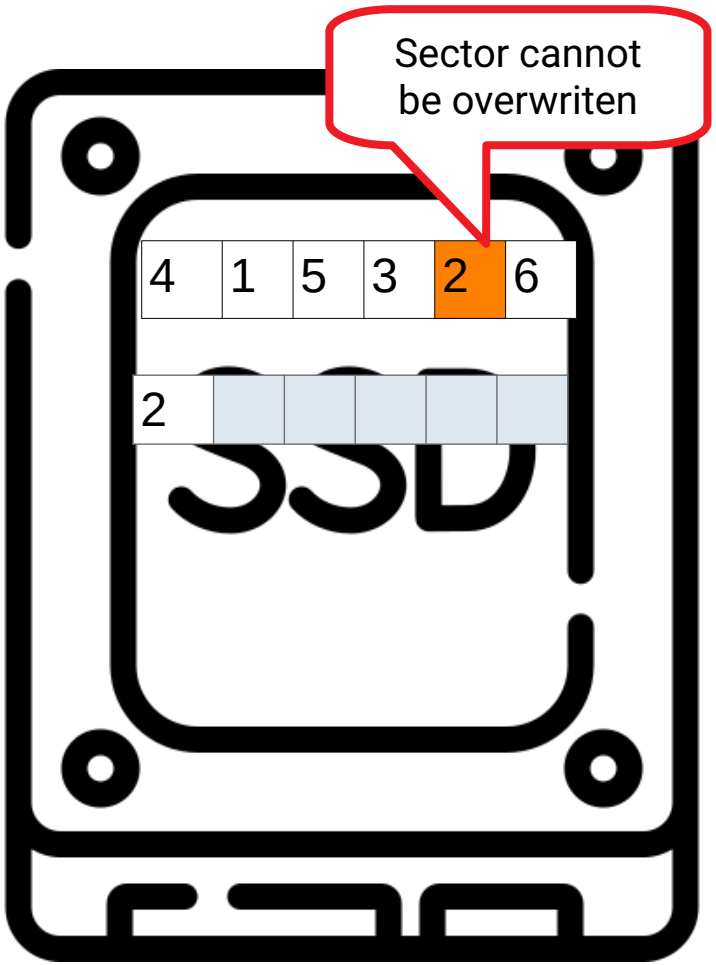
- Stack of multiple disk platters
- Mechanically seek a sector by:
  - Moving heads
  - Waiting for rotation until desired sector
- Read/write contigous sectors in a single rotation
- Cost and performance:
  - Low cost
  - High latency for random access
  - Higher heat generation and sensitivity
- Typical failure modes:
  - Individual sectors
  - Mechanical failure



Icons by Flaticon.com.

Solid state disks (SSD)

- Collection of FLASH memory chips
- Cannot overwrite single sector:
  - Erase a full block
  - Relocate sectors to an empty block on write/overwrite
- Read directly
- Benefits from TRIM operation:
  - Proactively erase blocks that are unused before they are overwritten
- Performance and host:
  - Low latency and high throughput
  - Higher cost and lower capacity
- Failure modes:
  - FLASH wear



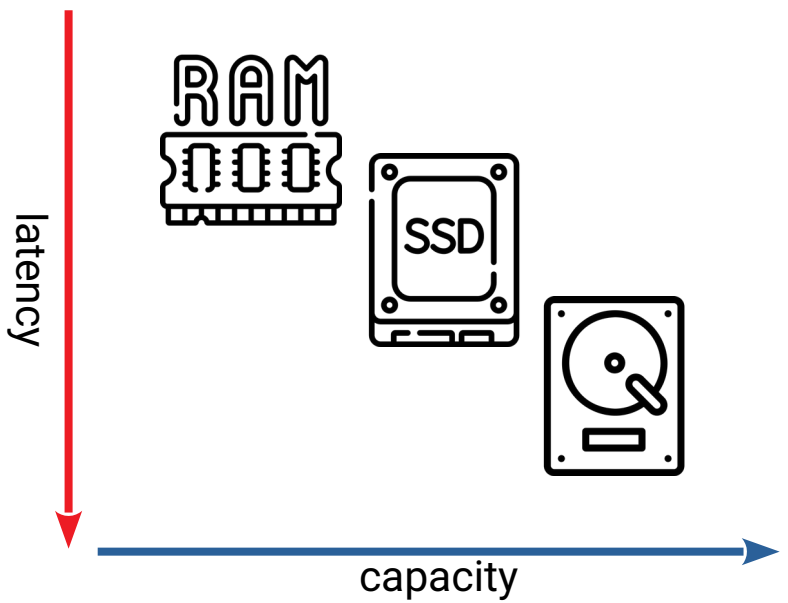
Icons by Flaticon.com.

Challenges

- Capacity
  - Not enough
  - Too much
  - Changing demand
- Performance
  - High latency
  - Low throughput
- Reliability

Caching and tiering

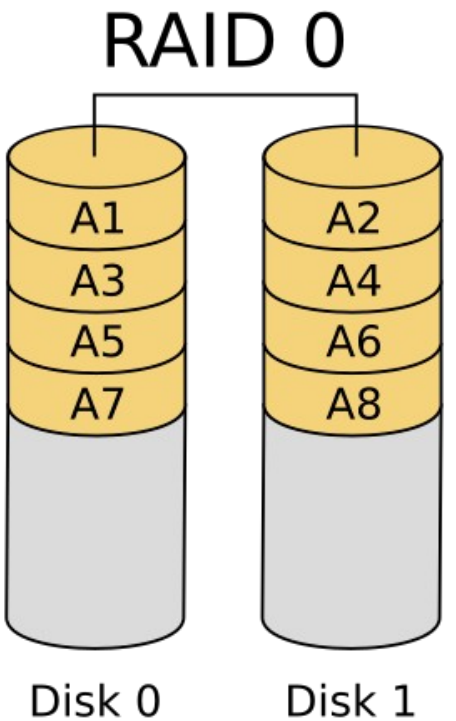
- Keep a subset of data in a smaller but high performance storage medium
- Requires access locality: revisiting recently used data
- Capacity
- Reliability
- Latency
- Throughput



Icons by Flaticon.com. Image from Wikipedia.

RAID

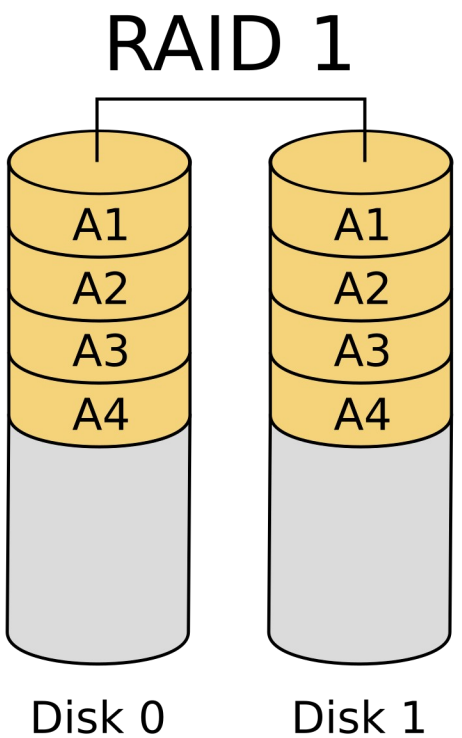
- **R**edundant **A**rray of **I**nexpensive **D**isks
- Level 0: Interleave sectors over more than one disk
- Capacity =
- Reliability ↓
- Latency ↑
- Throughput ↑



Icons by Flaticon.com. Image from Wikipedia.

RAID

- Level 1: Duplicate sectors over more than one disk
- Capacity
- Reliability
- Latency
- Throughput
  - Reading
  - Writing

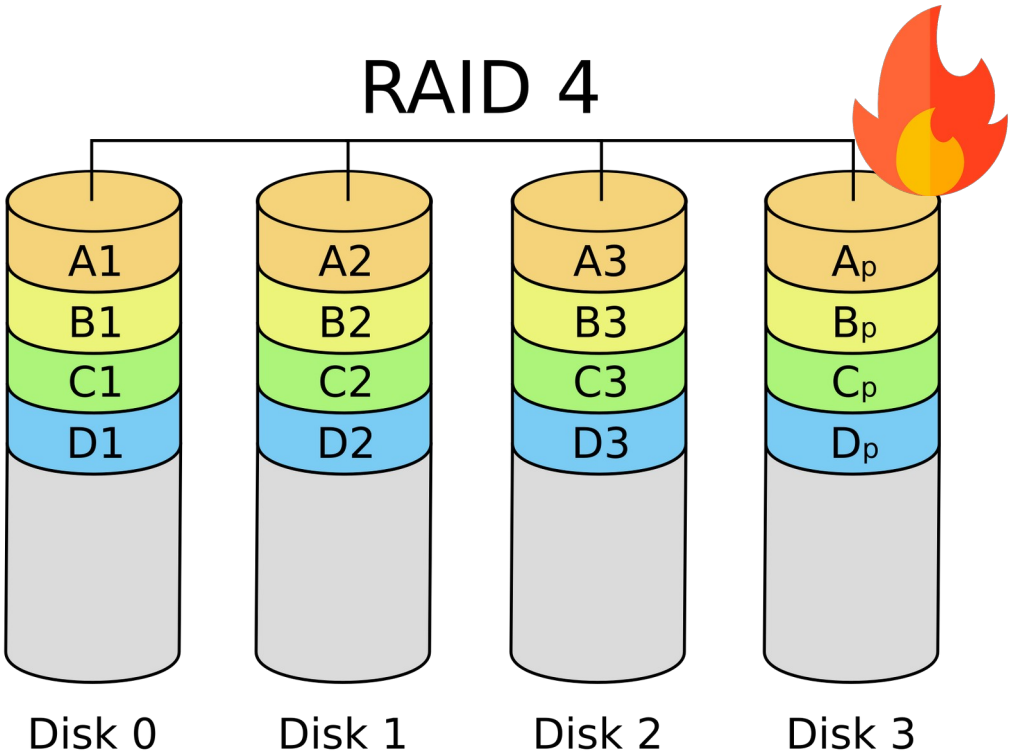


Icons by Flaticon.com. Image from Wikipedia.

RAID

- Level 4: Store parity in a dedicated disk
- Capacity
- Reliability
- Latency
- Throughput
  - Reading
  - Writing

- Must read others in same stripe
- Always writes to the same disk

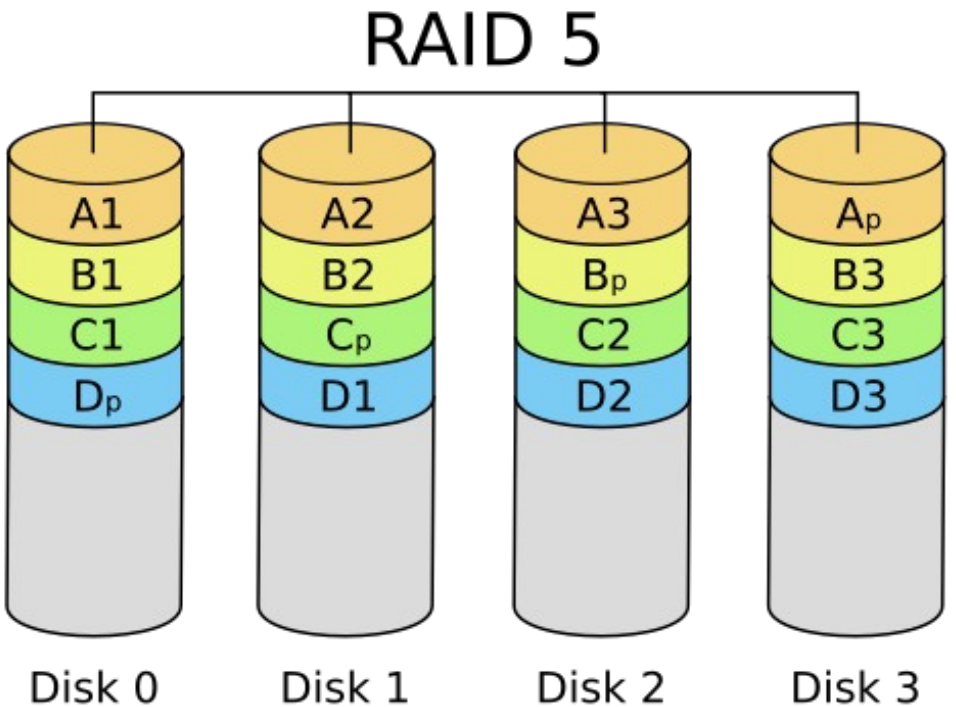


When writing: Ap = A1+A2+A3
When reading:
- normally, read A1
- upon failure: A1 = Ap-(A2+A3)

Icons by Flaticon.com. Image from Wikipedia.

RAID

- Level 5: Interleave parity in all disks
- Capacity
- Reliability
- Latency
- Throughput
  - Reading
  - Writing

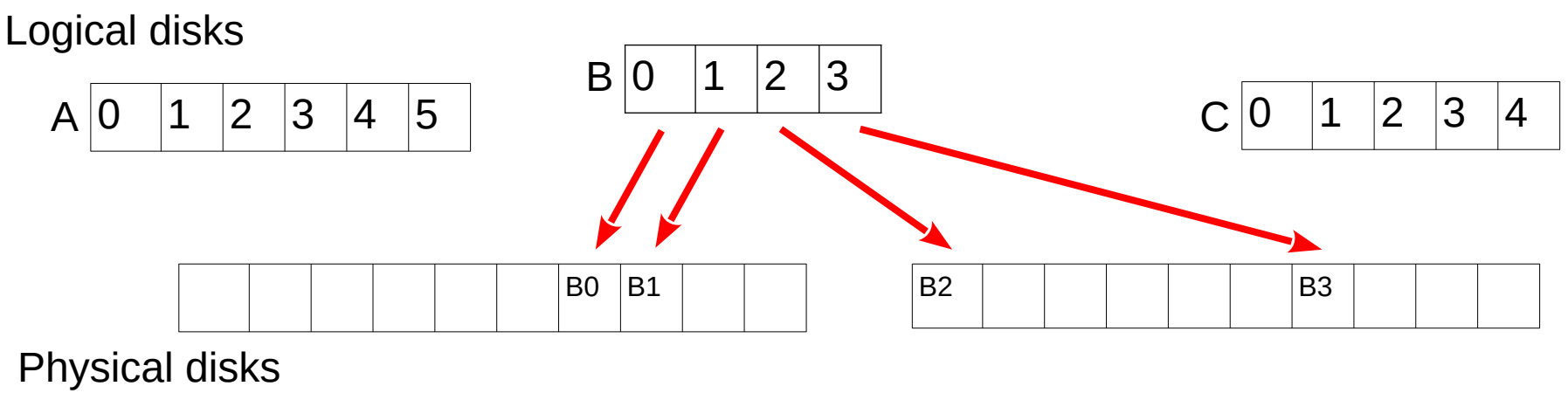


With effective caching, as no single disk is always updated

Icons by Flaticon.com. Image from Wikipedia.

Volume management

- Arbitrarily map physical disk extents to logical disks (volumes)
  - Allow dynamic reconfiguration



Volume management

- Can be provided over the network
  - Access “slices” of a large disk stored in a remote server



- Often combined with caching, tiering, and RAID

- Provided by cloud IaaS

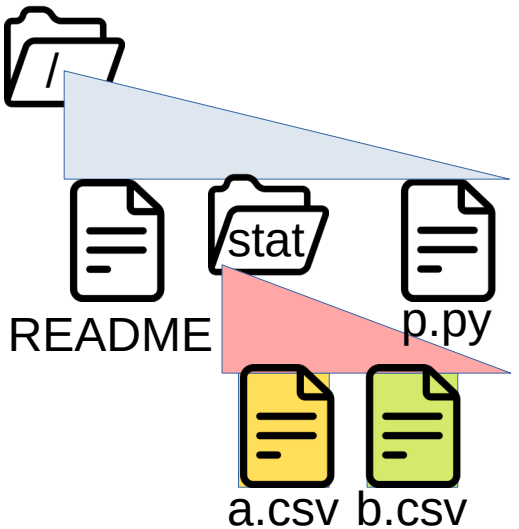
Summary

- Main challenges solved by arbitrary combinations of:
  - Caching
  - RAID levels
  - Volume management

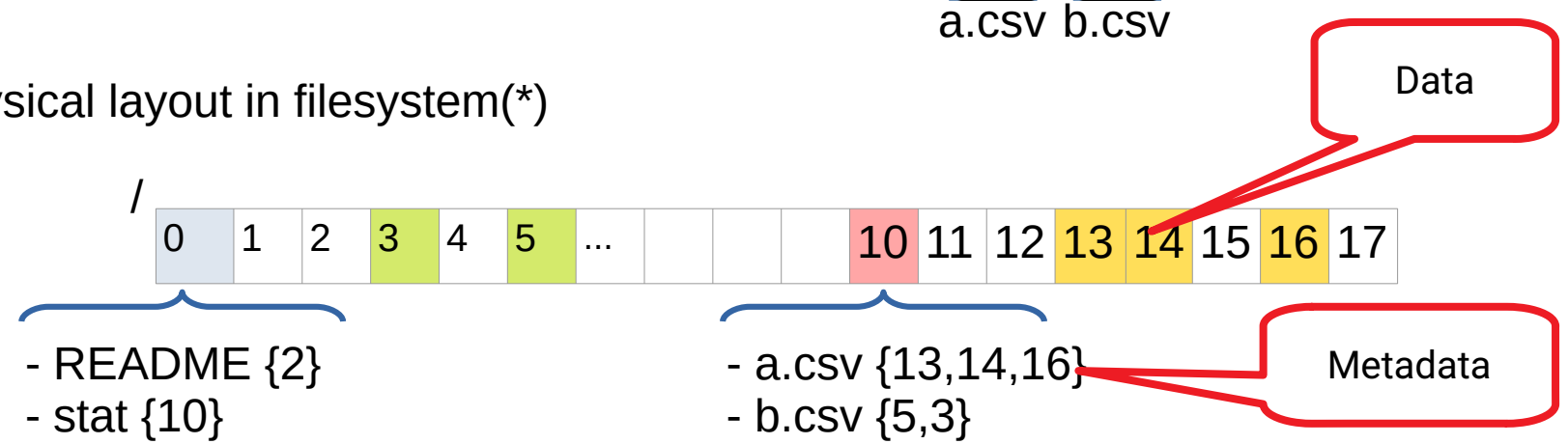
- Problem:
  - Sequences of fixed size blocks are not a good abstraction for applications and users

File systems

Logical organization of data



Physical layout in filesystem(*)



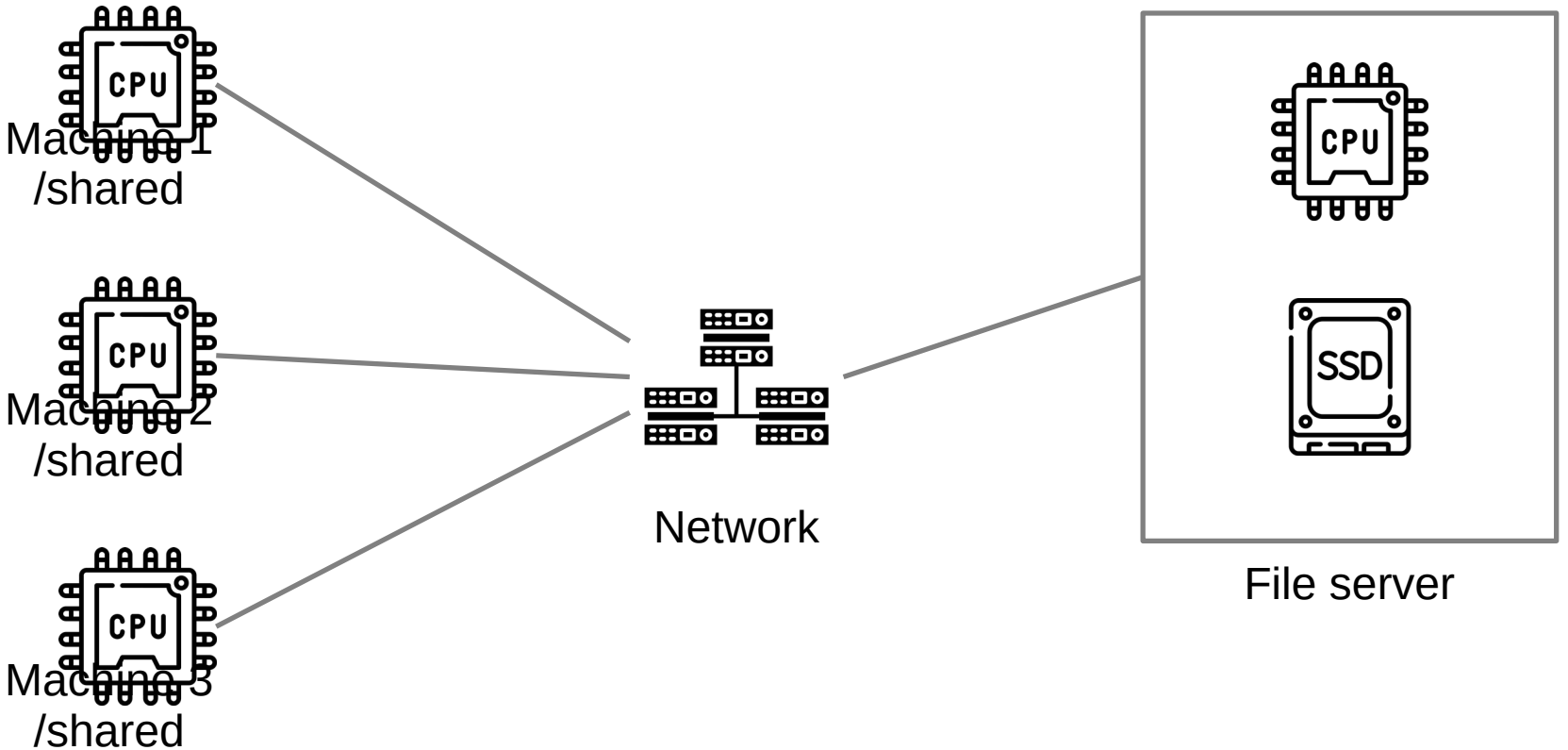
(*) Simplified view.

Icons by Flaticon.com.

File systems

- Additional funcionality:
  - Access control, compression, snapshotting, ...
- Sometimes include volume management functionality (zfs, btrfs, …)
- Can be “mounted”:
  - The folder hierarchy of a file system stored in a separate can be made to appear under a folder in a different file system

Network file systems



Network file systems

- The same interface, in another machine
  - Applications open, read, write, and seek as before
  - The hierarchy is mounted from a server over the network (NFS, SMB)
- Many clients at the same time
  - Several machines mount the same folder, for reading and writing
  - Neither a disk nor an object store allows this
- New failure modes and new costs
  - The network and the server, besides the disks
  - Every operation is a round trip
- Still a single server
  - Capacity and throughput are those of one machine
  - Managed services: Filestore, EFS, Azure Files

Scale

- What if data does not fit in disks attached to a single server?
  - Capacity
  - Throughput

- Design principles and goals for cloud storage:
  - Simplified abstraction (no hierarchy, append only, ...)
  - Separation of data and metadata
  - Cope with new failure modes: entire servers, DCs, ...
  - Decentralization and parallelism

Hadoop Distributed File System

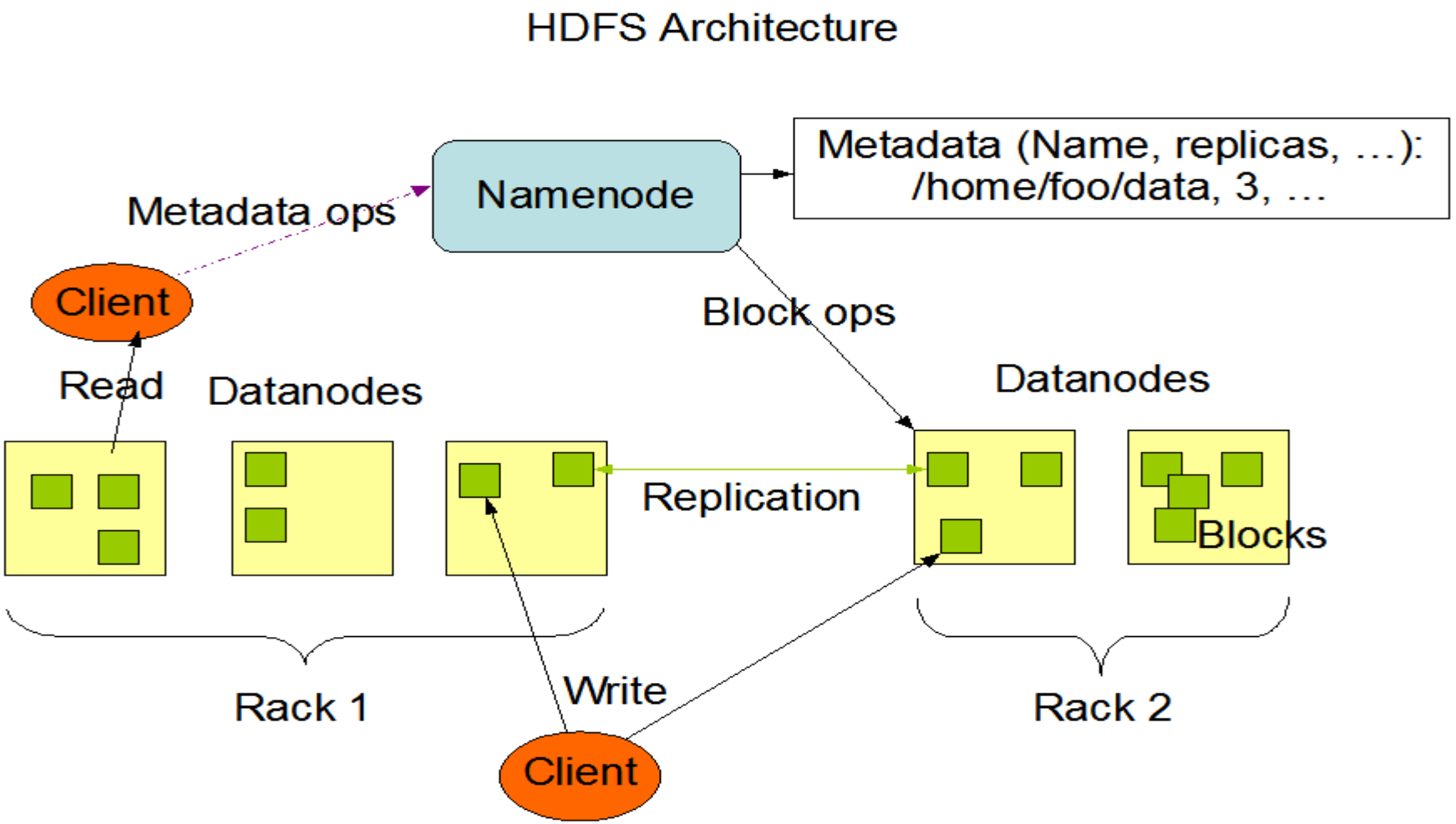


Image from: https://hadoop.apache.org/docs/current/hadoop-project-dist/hadoop-hdfs/HdfsDesign.html

Cloud object storage

Capacity and throughput	Amazon S3 holds more than **280 trillion objects** and averages over **100 million requests per second**
Events	Every day, Amazon S3 sends over **125 billion event notifications** to serverless applications
Replication	Customers use Amazon S3 Replication to **move more than 100 PB** of data per week
Cold Storage Retrieval	Every day, customers **restore more than 1PB** from the S3 Glacier Flexible Retrieval and S3 Glacier Deep Archive storage classes
Data Integrity Checks	Amazon S3 performs over **4 billion checksum computations per second**
Cost Optimization	On average, customers using Amazon S3 Storage Lens advanced metrics and recommendations have obtained **cost savings 6x greater** than the Storage Lens cost in the first six months of using it.
Flexibility	**Hundreds of thousands of data lakes** are built on Amazon S3

Fonte: “Building and operating a pretty big storage system called S3,” All Things Distributed, July 27, 2023.

https://www.allthingsdistributed.com/2023/07/building-and-operating-a-pretty-big-storage-system.html

Conclusion

- Cloud services encapsulate storage management concerns in two main abstractions
  - Networked volumes:
    - for program and private working area
  - Networked file systems:
    - for shared working areas
  - Object storage:
    - for long term data archival (*data lakes*)